








Advanced Time Series Analysis and Forecasting

Chapter 10: Long memory and rough volatility



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BSS**: Brownian Semistationary (process)
- **CPI**: Consumer Price Index
- **DM**: Diebold–Mariano (test)
- **ELW**: Exact Local Whittle (estimator)
- **EODHD**: EOD Historical Data (market-data provider)
- **FCVAR**: Fractionally Cointegrated VAR
- **FFT**: Fast Fourier Transform
- **FIGARCH**: Fractionally Integrated GARCH
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FTW**: Fukasawa, Takabatake and Westphal (2022)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GJR**: Gatheral, Jaisson and Rosenbaum (2018)
- **GMM**: Generalized Method of Moments
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)

Acronyms used in this material (2/2)

- **HICP**: Harmonised Index of Consumer Prices
- **HYGARCH**: Hyperbolic GARCH (Davidson)
- **LLM**: Large Language Model
- **LMSV**: Long-Memory Stochastic Volatility
- **LR**: Likelihood Ratio (test)
- **LW**: Local Whittle (estimator)
- **MA**: Moving Average
- **MFM**: Modelling Financial Markets
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **NBLS**: Narrow-Band Least Squares
- **OHLC**: Open, High, Low, Close
- **OLS**: Ordinary Least Squares
- **OU**: Ornstein–Uhlenbeck (process)
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **RFSV**: Rough Fractional Stochastic Volatility (model)
- **RMSE**: Root Mean Squared Error
- **RV**: Realised Variance
- **SD**: Standard Deviation
- **SE**: Standard Error
- **SV**: Stochastic Volatility
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model
- **VIX**: CBOE Volatility IndeX
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question**: how slowly do shocks to volatility and inflation die out, how rough is volatility at short horizons, and how do we tell genuine long memory from breaks and from measurement error?
  - two scales of the same object: the persistence of log volatility over months, its roughness over days
- **Route** of the chapter
  - spectral characterisation and fractional processes; asymptotics of GPH, local Whittle and exact local Whittle; bandwidth and bias
  - testing long memory against level shifts (Qu 2011); fractional cointegration (FCVAR)
  - long memory in volatility: FIGARCH, HYGARCH, LMSV, HAR as an approximation
  - rough volatility: fBm, simulation, the evidence of Gatheral, Jaisson and Rosenbaum, critiques, forecasting
- We build on TSA, Chapter 8 (ARFIMA, R/S, GPH), Chapter 1 (forecast evaluation), Chapter 2 (breaks) and Chapter 8 (realised measures, HAR); Seminar 10 comes before this lecture

Learning outcomes

- Characterise long memory in the time and frequency domains and derive the autocorrelations and spectrum of ARFIMA processes
- State the asymptotic theory of GPH, local Whittle and exact local Whittle, and choose the bandwidth from the bias–variance trade-off
- Test long memory against short memory with level shifts (Qu 2011) and estimate a fractionally cointegrated VAR
- Estimate FIGARCH and HYGARCH models, and long memory in volatility robust to measurement noise
- Simulate fractional Brownian motion, replicate the roughness evidence on realised variance, and evaluate rough-volatility forecasts honestly out of sample

Reading, data and tools

- Long memory: Granger and Joyeux (1980); Hosking (1981); Robinson (1995b); Shimotsu and Phillips (2005); Beran et al. (2013)
  - Breaks and co-memory: Qu (2011); Perron and Qu (2010); Johansen and Nielsen (2012)
  - Volatility: Baillie, Bollerslev and Mikkelsen (1996); Corsi (2009); Gatheral, Jaisson and Rosenbaum (2018); Bennedsen, Lunde and Pakkanen (2022)
- Python Quantlets of this chapter: Quantlets/Ch_10
  - periodogram estimators, the Qu test, FCVAR, FIGARCH, circulant embedding, the hybrid scheme and the RFSV predictor written in `numpy/scipy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source, sample	Use
Realised variance of six equity indices	Heber et al. (2009): 5-minute RV, January 2000 – February 2022 (end of the library)	memory, roughness, forecasts
Bitcoin realised variance	Binance one-minute prices (public data), our measures, January 2018 – September 2026	roughness, forecasts
S&P 500, DAX, BET, Bitcoin, VIX	EODHD daily OHLC and closes, 2000 – September 2026	FIGARCH, $\lvert r\rvert$, ranges, FCVAR
EUR/RON	BNR reference rate, July 2005 – September 2026	memory of $\lvert r\rvert$
Inflation	US CPI (FRED CPIAUCSL), Romanian HICP (Eurostat prc_hicp_minr), monthly	persistence and breaks

- Where no intraday data exist (BET, EUR/RON) volatility is proxied by $|r_t|$ and $\log r_t^2$; the Parkinson range is used to show how a noisier proxy distorts roughness

From the Nile to long memory

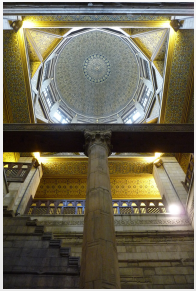


Harold Edwin Hurst, 1953

Photo: Elliott & Fry (1953); public domain; Wikimedia Commons

- Hurst studied centuries of Nile levels to size the reservoirs above Aswan Hurst (1951)
- The adjusted range of cumulated inflows grew like n^H with $H \approx 0.7$, not $n^{1/2}$: the **Hurst effect**
- Wet years follow wet years in long runs: dependence that no finite ARMA reproduces
- Two explanations followed
  - fractional Gaussian noise Mandelbrot and Van Ness (1968)
  - fractional differencing Granger and Joyeux (1980), Hosking (1981)

Records of the river

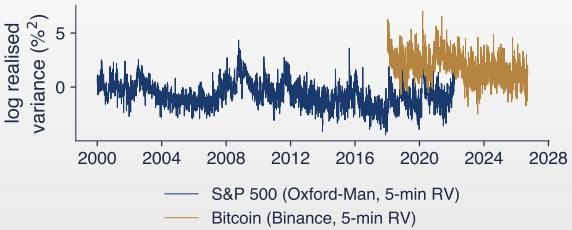


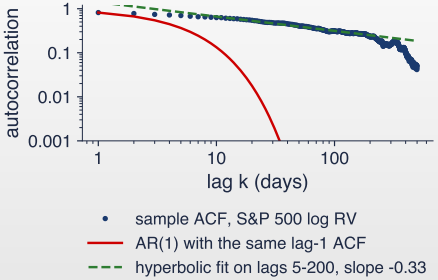
The Nilometer on Roda Island, Cairo

Photo: Willyman (2009); CC BY-SA 4.0; Wikimedia Commons

- The Roda Nilometer recorded annual Nile minima from the seventh century on: one of the oldest time series
- These minima are the classic long-memory data set Beran et al. (2013)
- Today the same question is asked of volatility, inflation and interest rates: does a shock fade geometrically or hyperbolically?
- And a newer one: at short horizons, is volatility smoother or rougher than Brownian motion?

Twenty-two years of log realised variance





⊡ Left: S&P 500 (Oxford-Man, $n = 5\,552$ days to 25 February 2022) and Bitcoin ($n = 3\,165$ days to 18 September 2026); right: sample ACF of S&P 500 log RV, lags 1–500, log-log

ATS_ch10_memory

Interpreting the long view

- ACF 0.82 at lag 1, 0.53 at lag 22, still 0.18 at lag 250: an AR(1) with the same first lag would give about 10^{-22} at lag 250
- On lags 5–200 the ACF is a straight line in log-log, slope -0.33: hyperbolic decay k^{2d-1} with $d \approx 0.34$
- The local Whittle estimate is larger, 0.62: the sample ACF is biased down at long lags under long memory
- The rest of the chapter asks which of these numbers to trust, and why the same series also looks rough

Known from TSA, and new here

- Known (TSA, Chapter 8): fractional differencing, ARFIMA(p, d, q), R/S and DFA, GPH and local Whittle as recipes, FIGARCH and HAR in brief, spurious long memory by example
- New: the theory behind the recipes
  - equivalent definitions, aggregation, the variance of the sample mean
  - limit laws of the semiparametric estimators, non-stationary d, bias and the choice of m
  - a formal test against level shifts; fractional cointegration; rough volatility
- Replications: Granger (1980) aggregation, Qu (2011) design, Gatheral, Jaisson and Rosenbaum (2018, Section 2) scaling on the Oxford-Man data they used

Two definitions of long memory (1/2)

- **Time domain**: the autocovariances decay hyperbolically and are not summable

$$\gamma(k) \sim c_\gamma \, k^{2d-1} \quad (k \to \infty), \quad 0 < d < 1/2 \;\Rightarrow\; \sum_k |\gamma(k)| = \infty$$

  - $\gamma(k)$: autocovariance at lag k; d: memory parameter; $c_\gamma > 0$: a constant; $a_k \sim b_k$ means $a_k / b_k \to 1$
  - short memory: $\sum_k |\gamma(k)| < \infty$ (e.g. ARMA, with geometric decay)
- **Frequency domain**: the spectral density has a pole at frequency zero

$$f(\lambda) \sim c_f \, \lambda^{-2d} \quad (\lambda \to 0^+)$$

  - $f(\lambda)$: spectral density at frequency $\lambda \in (0, \pi]$; $c_f > 0$: a constant
  - $d < 0$: antipersistence, $f(0) = 0$, $\sum_k \gamma(k) = 0$; $d = 0$: short memory

Two definitions of long memory (2/2)

- Equivalence: for quasi-monotone γ the constants are linked (Abelian–Tauberian theorems) Beran et al. (2013)

$$c_f = c_\gamma \, \Gamma(2d) \, \frac{\sin(\pi/2 - \pi d)}{\pi}$$

  - $\Gamma(\cdot)$: the gamma function; without such regularity the two definitions differ
  - semiparametric estimators use only the frequency-domain definition
- Hurst scale: $H = d + 1/2$ for the stationary series
  - $H > 1/2$: persistent; $H < 1/2$: antipersistent; $H = 1/2$: short memory

Fractional integration and ARFIMA (1/2)

- The fractional difference operator is a power series in the lag operator L ($LX_t = X_{t-1}$) Granger and Joyeux (1980), Hosking (1981)

$$(1-L)^d = \sum_{k\geq 0} \pi_k L^k, \qquad \pi_0 = 1, \quad \pi_k = \pi_{k-1}\frac{k-1-d}{k}, \quad \pi_k \sim \frac{k^{-d-1}}{\Gamma(-d)}$$

  - π_k: the weight of X_{t-k}; it decays hyperbolically, so the filter has infinite memory; $d = 1$ gives the usual difference $1 - L$

- ARFIMA(p, d, q): an ARMA for the fractionally differenced series

$$\phi(L)(1-L)^d(X_t - \mu) = \theta(L)\varepsilon_t$$

  - $\phi(L)$, $\theta(L)$: AR and MA polynomials of orders p, q; μ: mean; ε_t: white noise with variance σ^2
  - stationary and invertible for $-1/2 < d < 1/2$

Fractional integration and ARFIMA (2/2)

- Spectrum: a pole at zero times the spectrum of the ARMA part

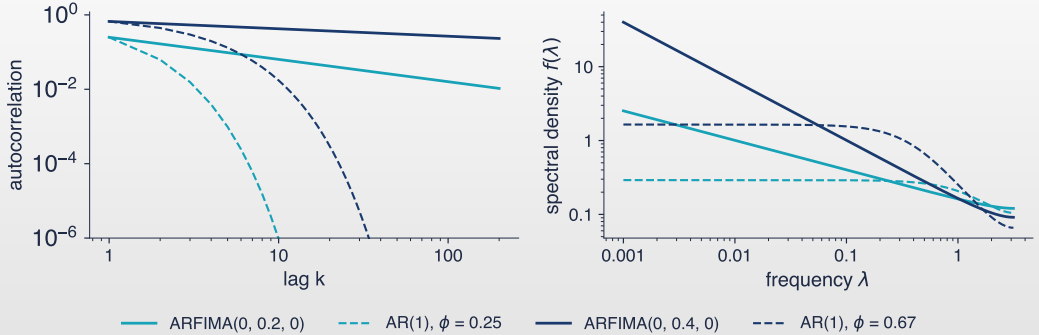
$$f(\lambda) = \frac{\sigma^2}{2\pi} \left|2\sin(\lambda/2)\right|^{-2d} \frac{|\theta(e^{-i\lambda})|^2}{|\phi(e^{-i\lambda})|^2} = |2\sin(\lambda/2)|^{-2d} f^*(\lambda)$$

  - f^*: the smooth short-memory (ARMA) part; near zero $|2\sin(\lambda/2)| \approx \lambda$, so $f(\lambda) \approx f^*(0)\lambda^{-2d}$
- ARFIMA$(0, d, 0)$: variance and autocorrelations in closed form

$$\gamma(0) = \sigma^2 \frac{\Gamma(1-2d)}{\Gamma(1-d)^2}, \qquad \rho(k) = \frac{\Gamma(k+d)\Gamma(1-d)}{\Gamma(k-d+1)\Gamma(d)} \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$$

  - recursion $\rho(k) = \rho(k-1)\,(k-1+d)/(k-d)$, $\rho(1) = d/(1-d)$

Hyperbolic and geometric decay



- Left: ACF of ARFIMA$(0, d, 0)$ and of an AR(1) with the same $\rho(1)$, log-log; right: spectral densities near zero, the AR(1) scaled to the same variance

ATS_ch10_memory

Interpreting the two decays

- $d = 0.4$: $\rho(1) = 0.667$, $\rho(10) = 0.424$, $\rho(100) = 0.267$; the matching AR(1) has $\rho(100)$ of order 10^{-18}
- Partial sums $\sum_{k \le 100} \rho(k) = 33.0$ and $\sum_{k \le 200} \rho(k) = 57.8$: they keep growing, like K^{2d}
- In the spectrum the difference is a pole against a plateau: long memory lives at the lowest frequencies, where few periodogram ordinates exist
- Hence all semiparametric estimators look only at the m lowest Fourier frequencies

Where long memory comes from: aggregation

- Granger (1980): the scaled sum of N independent AR(1) series with random coefficients

$$X_t = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} x_{it}, \qquad x_{it} = \phi_i x_{i,t-1} + \varepsilon_{it}, \qquad \phi_i^2 \sim \mathrm{Beta}(p, q)$$

  - ϕ_i: persistence of unit i, drawn once; Beta density $\propto \phi^{2p-1}(1-\phi^2)^{q-1}$ on $(0,1)$; q controls the mass near the unit root

- Derivation: average the AR(1) autocovariances over the distribution of ϕ

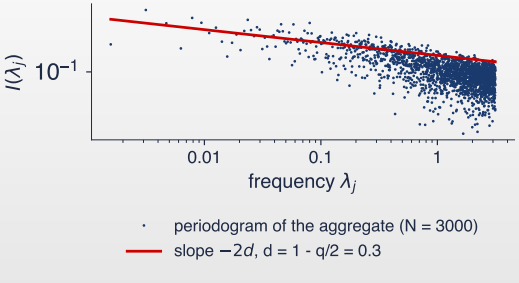
$$\gamma_X(k) = \mathbb{E}\Big[\frac{\phi^k}{1-\phi^2}\Big] \propto \int_0^1 \phi^{k+2p-1}(1-\phi^2)^{q-2} d\phi \propto B\Big(\frac{k}{2}+p, q-1\Big) \sim c\, k^{-(q-1)}$$

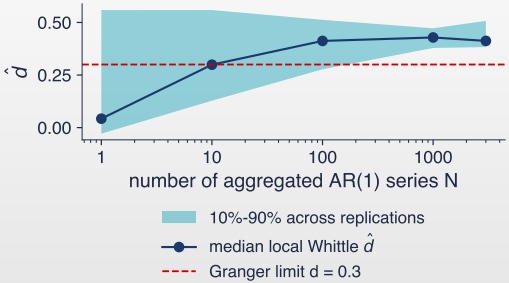
  - $B(\cdot,\cdot)$: the beta function; matching k^{2d-1} gives $d = 1 - q/2$, long memory for $1 < q < 2$

- Economic reading: aggregate inflation, volatility or output mixes many short-memory agents with heterogeneous persistence; a few near-unit-root units dominate the low frequencies

- Long memory can therefore be structural, not a curiosity of the estimator

Aggregation of AR(1) series





- $\phi_i^2 \sim \mathrm{Beta}(1, 1.4)$, so $d = 1 - q/2 = 0.3$; $n = 4000$; local Whittle with $m = 219$, 20 replications per N; left: periodogram of one aggregate of 3000 series

ATS_ch10_memory

Interpreting aggregation

- One AR(1): median $\hat{d} = 0.04$; ten series: 0.30; 100 series: 0.41; 3000 series: 0.41 (10%–90%: 0.38 to 0.51)
- The estimate settles above the limit 0.3: the aggregate spectrum is λ^{-2d} only very close to zero, and $m = n^{0.65}$ reaches into the region where it is not
- This is the bias problem of Section 2 in its purest form: a semiparametric estimator is only as good as its bandwidth
- The direction of the effect, from no memory to clear long memory, is the point of Granger's theorem

Consequences for inference: the sample mean

- The variance of the sample mean $\bar{X}_n$ decays more slowly than $1/n$

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n}\sum_{|k|<n}\Big(1-\frac{|k|}{n}\Big)\gamma(k) \sim \frac{c_\gamma}{d(2d+1)}\, n^{2d-1}$$

  - the mean converges at rate $n^{1/2-d}$, not $n^{1/2}$; with $d = 0.4$: $n^{0.1}$
- HAC standard errors (Chapter 0) assume $\sum_k |\gamma(k)| < \infty$: under long memory the long-run variance is infinite and Newey–West intervals are too narrow
  - the same holds for DM tests (Chapter 1) on loss differences that inherit long memory
- Regression with long-memory regressors and errors: OLS slopes converge slowly
  - spurious regression between independent fractional series when $d_x + d_u > 1/2$ (d_x, d_u: memory of the regressor and of the error)
- Every test of this course built on $\sqrt{n}$ asymptotics needs a second look when $d > 0$

Recap: Long memory

- Long memory: hyperbolic ACF k^{2d-1} and a spectral pole λ^{-2d}; $H = d + 1/2$
- ARFIMA separates the pole $|2\sin(\lambda/2)|^{-2d}$ from the short-memory part f^*
- Aggregation of heterogeneous AR(1) units produces long memory; the mean converges at $n^{1/2-d}$

The periodogram near frequency zero

- The periodogram: the squared modulus of the discrete Fourier transform, at the Fourier frequencies

$$I(\lambda_j) = \frac{1}{2\pi n}\left|\sum_{t=1}^{n} X_t e^{-i\lambda_j t}\right|^2, \qquad \lambda_j = \frac{2\pi j}{n}, \quad j = 1, \ldots, m$$

  - n: sample size; m: bandwidth, the number of lowest frequencies used; short memory: $I(\lambda_j)/f(\lambda_j)$ asymptotically i.i.d. $\mathrm{Exp}(1)$
- Long memory: for fixed j, $\mathbb{E}\, I(\lambda_j)/f(\lambda_j) \not\to 1$ and the ordinates stay correlated Robinson (1995a)
  - the distortion vanishes as $j \to \infty$: the proofs need $m \to \infty$, $m/n \to 0$, and handle the first frequencies separately
- Semiparametric model: only the behaviour near zero is specified

$$f(\lambda) = G\lambda^{-2d}\left(1 + O(\lambda^{\beta})\right), \qquad \lambda \to 0, \quad \beta \in (0, 2]$$

  - $G > 0$: level of the spectrum at low frequencies; β: smoothness of the remainder; $\beta = 2$ for ARFIMA, because f^* is smooth and even

GPH: the log-periodogram regression

- Taking logs of $I = f\xi$ gives a linear regression in $\log \lambda_j$ with slope $-2d$ Geweke and Porter-Hudak (1983)

$$\log I(\lambda_j) = \log G - 2d \log \lambda_j + \log \xi_j, \qquad \xi_j = \frac{I(\lambda_j)}{f(\lambda_j)}$$

  - OLS on $j = 1, \dots, m$; with $\xi_j \sim \mathrm{Exp}(1)$: $\mathbb{E} \log \xi_j = -\gamma_E$ ($\gamma_E \approx 0.577$, Euler's constant), $\mathrm{Var} \log \xi_j = \pi^2/6$
- Robinson (1995a): for $|d| < 1/2$ and Gaussian X, if $m \to \infty$ and $m^5/n^4 \to 0$

$$\sqrt{m}(\hat{d}_{GPH} - d) \to N(0, \pi^2/24)$$

  - origin of $\pi^2/24$: $\mathrm{Var}(\hat{d}) = \dfrac{\pi^2/6}{4\sum_j \nu_j^2}$, with $\nu_j = \log j - \overline{\log j}$ the centred regressor and $\sum_j \nu_j^2 \sim m$
- Bias of order $(m/n)^2$ (formula on the bandwidth slide below)
  - the condition $m^5/n^4 \to 0$ makes the bias negligible relative to the standard deviation $m^{-1/2}$

Local Whittle (1/2): the objective

- Whittle approximation of the Gaussian log-likelihood, restricted to $j \le m$, with $f(\lambda_j) = G\lambda_j^{-2d}$

$$Q(G, d) = \frac{1}{m} \sum_{j=1}^{m} \left[\log \left(G\lambda_j^{-2d} \right) + \frac{I(\lambda_j)}{G\lambda_j^{-2d}} \right]$$

  - the same form as QLIKE (Chapter 8): the model spectrum $G\lambda_j^{-2d}$ is fitted to the periodogram ordinate by ordinate
- Concentrate out G Robinson (1995b): for given d the optimal level is the average of $\lambda_j^{2d} I(\lambda_j)$

$$\hat{G}(d) = \frac{1}{m} \sum_{j} \lambda_j^{2d} I(\lambda_j), \qquad \hat{d}_{LW} = \arg\min_{d} R(d)$$

$$R(d) = \log \hat{G}(d) - \frac{2d}{m} \sum_{j=1}^{m} \log \lambda_j$$

  - a one-dimensional minimisation in d

Local Whittle (2/2): the variance

- Score at the truth: a weighted sum of $\xi_j - 1$

$$R'(d) = \frac{2}{m} \sum_j \nu_j \Big(\frac{\lambda_j^{2d} I(\lambda_j)}{\hat{G}(d)} - 1 \Big)$$

  - $\nu_j = \log \lambda_j - \overline{\log \lambda}$: centred log frequency; with $\mathrm{Var}(\xi_j) = 1$ and $R''(d) \to 4$: $\sqrt{m}(\hat{d} - d) \to N(0, 4/16) = N(0, 1/4)$
- No distributional assumption beyond a linear process with martingale-difference innovations
  - efficiency relative to GPH: $(\pi^2/24)/(1/4) = \pi^2/6 \approx 1.64$
  - derivation step by step: Appendix

Local Whittle: the theorem and its limits

- Robinson (1995b): for $d \in (-1/2, 1/2)$, $f(\lambda) = G\lambda^{-2d}(1 + O(\lambda^{\beta}))$ and a bandwidth with $1/m + m^{1+2\beta}(\log m)^2/n^{2\beta} \to 0$
  - $\hat{d} \to_p d$ and $\sqrt{m}(\hat{d} - d) \to N(0, 1/4)$; the condition on m: it grows, but slowly enough for the bias to vanish
- Non-stationary d: consistent for $d < 1$, asymptotically Normal for $d < 3/4$ Velasco (1999)
  - for $d > 1$ it converges in probability to 1; between $3/4$ and 1 the limit is non-Normal Phillips and Shimotsu (2004)
- Remedies: difference the data first (needs knowing $d > 1/2$), taper the periodogram (variance cost), or estimate exactly

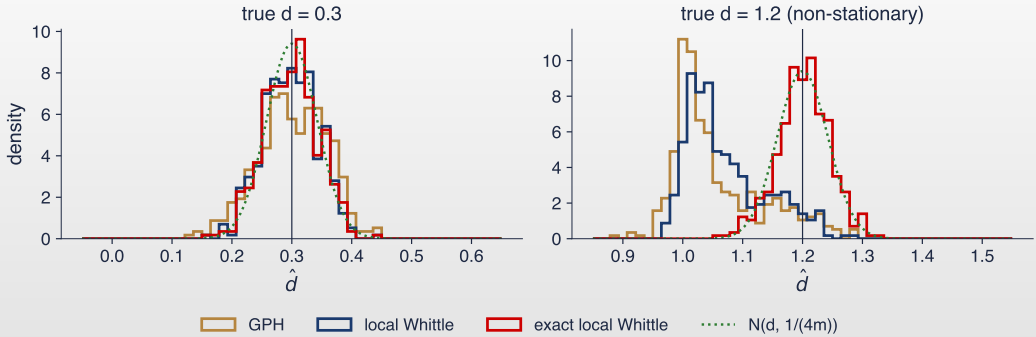
Exact local Whittle

- Shimotsu and Phillips (2005): replace $\lambda_j^{2d} I_X(\lambda_j)$ by the periodogram of the fractionally differenced data

$$\Delta^d X_t = \sum_{k=0}^{t-1} \pi_k X_{t-k}, \qquad R_E(d) = \log\Big(\frac{1}{m}\sum_j I_{\Delta^d X}(\lambda_j)\Big) - \frac{2d}{m}\sum_j \log \lambda_j$$

  - $\Delta^d = (1-L)^d$ truncated at the start of the sample; $I_{\Delta^d X}$: periodogram of the differenced series
  - $\lambda^{2d} I_X(\lambda)$ approximates $I_{\Delta^d X}(\lambda)$ only for $d < 1/2$; the exact version removes the error that breaks LW for large d: $\sqrt{m}(\hat{d}_{ELW} - d) \to N(0, 1/4)$ on any search interval of width $< 9/2$
- Unknown mean Shimotsu (2010): subtract $\hat{\mu}(d) = w(d)\bar{X} + (1 - w(d))X_1$
  - $w(d) = 1$ for $d \le 1/2$, 0 for $d \ge 3/4$, smooth in between: the sample mean is a bad estimator of the level when $d > 1/2$
- Cost: one fractional difference per evaluation, $O(n \log n)$ by FFT

GPH, local Whittle and exact local Whittle by Monte Carlo



- 400 Gaussian ARFIMA$(0, d, 0)$ paths, $n = 2000$, $m = n^{0.65} = 139$; asymptotic SD: GPH 0.054, local Whittle 0.042; $d = 1.2$ is simulated as the partial sum of an ARFIMA$(0, 0.2, 0)$

ATS_ch10_estimation

Interpreting the Monte Carlo

- $d = 0.3$: means 0.301, 0.297, 0.298; SD 0.057, 0.045, 0.045; 95% coverage 94%, 94%, 95% (GPH, LW, ELW)
- The asymptotic laws are accurate at $n = 2000$; local Whittle is about $\sqrt{\pi^2/6}$ times tighter than GPH
- $d = 1.2$: GPH 1.057 and LW 1.072 collapse towards 1 (coverage 26% and 22%); ELW 1.202, coverage 93%
- For series that may be non-stationary (prices, inflation in levels, log RV with d near 0.5) report ELW

Bandwidth: bias against variance

- Leading bias (GPH and LW) Hurvich, Deo and Brodsky (1998): the curvature of $\log f^*$ near zero leaks into the slope

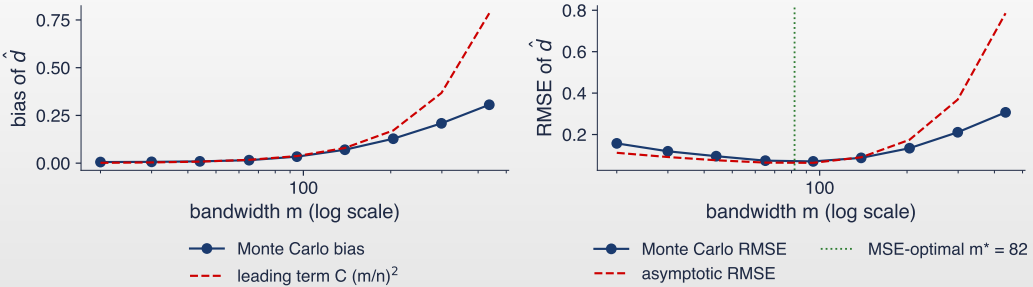
$$\mathbb{E}\hat{d} - d \approx -\frac{2\pi^2}{9}\,\frac{f^{*\prime\prime}(0)}{f^*(0)}\Big(\frac{m}{n}\Big)^2 = C\Big(\frac{m}{n}\Big)^2$$

  - from regressing $\log f^*(\lambda_j) \approx \log f^*(0) + b\lambda_j^2$, $b = \tfrac{1}{2}f^{*\prime\prime}(0)/f^*(0)$, on $-2\nu_j$; C: the bias constant
  - ARFIMA$(1,d,0)$: $f^{*\prime\prime}(0)/f^*(0) = -2\phi/(1-\phi)^2$, upward bias for $\phi > 0$
- MSE: squared bias plus variance; minimising over m gives the optimal bandwidth Henry and Robinson (1996)

$$\mathrm{MSE}(m) \approx C^2\Big(\frac{m}{n}\Big)^4 + \frac{1}{4m}, \qquad m^* = \Big(\frac{n^4}{16C^2}\Big)^{1/5} \propto n^{4/5}$$

  - C is unknown: plug-in rules estimate it; local polynomial Whittle Andrews and Sun (2004) removes the λ^2 term at a variance cost
- At m^* bias and SD are of the same order, so nominal intervals undercover; practice: plot $\hat{d}(m)$ over a range and report it with the intervals

Bias and RMSE across bandwidths



- Local Whittle, ARFIMA$(1, 0.3, 0)$ with $\phi = 0.6$, $n = 2000$, 300 replications, $m = n^a$ for $a = 0.40, \ldots, 0.80$; dashed: the leading bias $C(m/n)^2$ with $C = 16.4$ and the asymptotic RMSE

ATS_ch10_estimation

Interpreting the bandwidth trade-off

- Small m ($a = 0.40$): bias 0.006, RMSE 0.16, dominated by variance; large m ($a = 0.80$): bias 0.31, RMSE 0.31
- Smallest Monte Carlo RMSE 0.070 at $m = 95$ ($a = 0.60$); theory: $m^* = 82$ ($a = 0.58$)
- The leading term overstates the bias at large m (0.79 against 0.31): the λ^2 expansion is local
- A fixed rule such as $m = n^{0.8}$ can turn a short-memory AR component into apparent long memory

Parametric alternative: Whittle and exact ML

- Whittle: fit a fully specified spectral density over all frequencies Fox and Taqqu (1986)

$$\hat{\vartheta} = \arg\min_{\vartheta} \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} \left[\log f(\lambda_j; \vartheta) + \frac{I(\lambda_j)}{f(\lambda_j; \vartheta)} \right]$$

  - ϑ: all ARFIMA(p, d, q) parameters (d, the AR and MA coefficients, σ^2); $\sqrt{n}$-consistent and efficient if the model is right; $\mathrm{Var}(\hat{d}) \approx 6/(\pi^2 n)$ for $(0, d, 0)$

- The trade-off is robustness against efficiency: a wrong (p, q) biases $\hat{d}$ for every n
  - semiparametric: rate $\sqrt{m}$, robust to f^*; parametric: rate $\sqrt{n}$, exposed to misspecification

- Common practice: semiparametric $\hat{d}$ first, then a parametric model whose d falls inside its interval, then forecasts

Recap: Semiparametric estimation

- GPH: $N(d, \pi^2/(24m))$; local Whittle: $N(d, 1/(4m))$; exact local Whittle keeps $N(d, 1/(4m))$ for non-stationary d
- Bias $\propto (m/n)^2$, SD $\propto m^{-1/2}$: MSE-optimal $m \propto n^{4/5}$, with an unknown constant
- Always report $\hat{d}(m)$ across bandwidths

Mechanisms of spurious long memory

- Diebold and Inoue (2001): Markov switching in the mean with switching probability $p_n \to 0$ as $n \to \infty$ produces $\mathrm{Var}(\sum_{t \le n} X_t) = O(n^{2d+1})$, like an I($d$) process
  - rare breaks are invisible at high frequencies and look like a pole at low ones Granger and Hyung (2004)
- Random level shifts Lu and Perron (2010): a short-memory series u_t around a level that jumps rarely

$$X_t = \mu_t + u_t, \qquad \mu_t = \mu_{t-1} + \delta_t \eta_t, \qquad \delta_t \sim \mathrm{Bernoulli}(p)$$

  - δ_t: 1 on the (rare) days with a shift; η_t: shift size, variance σ_η^2; spectrum $\approx p\sigma_\eta^2/(2\pi\lambda^2)$ above frequency $\approx p$, plus the flat f_u: a slope -2 that fades into a plateau
- Signature Perron and Qu (2010): under level shifts $\hat{d}(m)$ falls steeply as m grows; under true long memory it is flat
  - the same reasoning applies to trends, regime switches and deterministic seasonality left in the data
- TSA, Chapter 8 showed the phenomenon by example; here: a formal test with known size

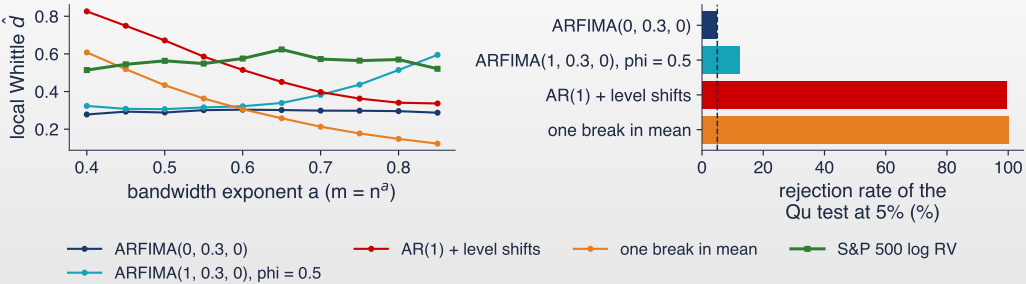
The Qu (2011) test

- Under H_0 (stationary long memory) the LW score has mean zero in every frequency band; under level shifts the low frequencies are over-weighted Qu (2011)

$$W = \sup_{r \in [\varepsilon, 1]} \Big(\sum_{j=1}^{m} \nu_j^2 \Big)^{-1/2} \Big| \sum_{j=1}^{\lfloor mr \rfloor} \nu_j \Big(\frac{I(\lambda_j)}{\hat{G} \lambda_j^{-2\hat{d}}} - 1 \Big) \Big|$$

  - $\hat{d}, \hat{G}$: local Whittle on the same m; $\nu_j = \log \lambda_j - \overline{\log \lambda}$; r: the share of the m frequencies included; ε: trimming of the lowest band
  - the partial sum at $r = 1$ is zero by the first-order condition; W: the largest standardised partial sum
- Limit: the supremum of a Gaussian process free of d and G; Qu recommends $m = n^{0.7}$, $\varepsilon = 0.02$ (large n) or 0.05
  - our simulated critical values: $\varepsilon = 0.02$: 1.09, 1.22, 1.56; $\varepsilon = 0.05$: 1.00, 1.14, 1.45 (10%, 5%, 1%)
  - reject for large W: evidence against stationary long memory, typically breaks, level shifts or trends

Size, power and the bandwidth signature



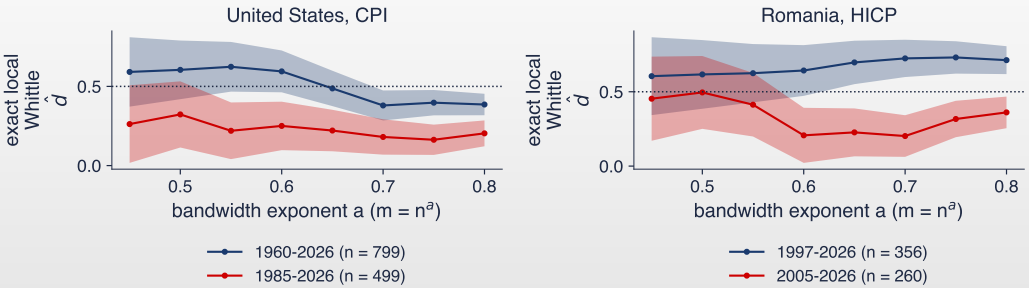
- $n = 2000$, 300 replications, $m = n^{0.7} = 204$, $\varepsilon = 0.02$; left: mean local Whittle $\hat{d}(m)$ (100 paths) and the S&P 500 log RV; right: rejection rates at 5%

ATS_ch10_qu_test

Interpreting the Qu test

- Size: 5% under ARFIMA$(0, 0.3, 0)$, 12% when an AR(1) with $\phi = 0.5$ is added: short-run dynamics distort size in finite samples
- Power: 99% against level shifts and 100% against one break, although LW reports $\hat{d} = 0.39$ and 0.21
- Signature: under level shifts $\hat{d}(m)$ drops from 0.83 to 0.34; S&P 500 log RV stays between 0.51 and 0.62, and $W = 0.66$ is far below the 10% value
- The memory of realised variance survives this test; Section 5 shows that $|r_t|$ of the BET and EUR/RON does not

Inflation persistence and the disinflation



- Monthly inflation, % annualised: US CPI (seasonally adjusted) to August 2026, Romanian HICP (monthly means removed) to August 2026; exact local Whittle with unknown mean, 95% bands; full sample and after the disinflation (US from 1985, Romania from 2005)

ATS_ch10_qu_test

Interpreting inflation persistence

- US: $\hat{d} = 0.49$ (SE 0.06) on 1960–2026 but 0.22 after 1985; Qu $W = 1.62$ (1% value 1.45) rejects for the full sample, $W = 0.55$ after 1985
- Romania: $\hat{d} = 0.70$ with the 1997–2004 disinflation, 0.23 from 2005 (range 0.20 to 0.50 across bandwidths); $W = 0.86$ and 1.04 with only 356 and 260 months
- Plain local Whittle on the full Romanian sample gives 0.31: with a trend-like level shift only the exact version with the Shimotsu mean is reliable
- Within a stable monetary regime inflation keeps moderate long memory, $\hat{d} = 0.22$ and 0.23 Hassler and Wolters (1995), Baillie, Chung and Tieslau (1996); the large estimates belong to regime changes

Recap: Short or long memory

- Breaks, level shifts and rare regime switches create a spectral pole and a large $\hat{d}$
- Diagnostic: $\hat{d}(m)$ falling with m; test: Qu (2011), with simulated critical values
- US inflation: long memory is mostly the Great Inflation; log RV passes the test

Fractional cointegration and the FCVAR (1/2)

- $X_t \in \mathbb{R}^p$ is I(d); it is **fractionally cointegrated** if $\beta' X_t$ is I($d-b$), $b > 0$
  - d: memory of the series; b: reduction of memory achieved by the combination $\beta' X_t$; Chapter 4 is the case $d = b = 1$
- FCVAR Johansen and Nielsen (2012): the VECM with the difference Δ replaced by fractional differences

$$\Delta^d X_t = \alpha\beta' L_b \Delta^{d-b} X_t + \sum_{i=1}^{k} \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t, \qquad L_b = 1 - \Delta^b$$

  - $\Delta^d = (1-L)^d$; L_b: fractional lag operator ($L_1 = L$), so $d = b = 1$ gives the VECM
  - α ($p \times r$): adjustment; β ($p \times r$): cointegrating vectors, rank r; Γ_i: short-run matrices, k lags; ε_t: i.i.d. errors

Fractional cointegration and the FCVAR (2/2): estimation and rank

- For fixed (d, b): a reduced-rank regression of $Z_0 = \Delta^d X$ on $Z_1 = L_b \Delta^{d-b} X$; then profile the likelihood over (d, b)

$$\ell_r(d, b) = -\frac{T}{2}\Big[\log\det S_{00} + \sum_{i \le r} \log(1 - \hat{\lambda}_i)\Big]$$

  - S_{00}: residual covariance of Z_0 (after the short-run terms); $\hat{\lambda}_i$: the Johansen eigenvalues, in decreasing order; T: sample size
- Rank tests: the LR trace statistic for rank r against p
  - for $b < 1/2$ it is asymptotically $\chi^2_{(p-r)^2}$; for $b > 1/2$ use the fractional Dickey–Fuller distributions of MacKinnon and Nielsen (2014)

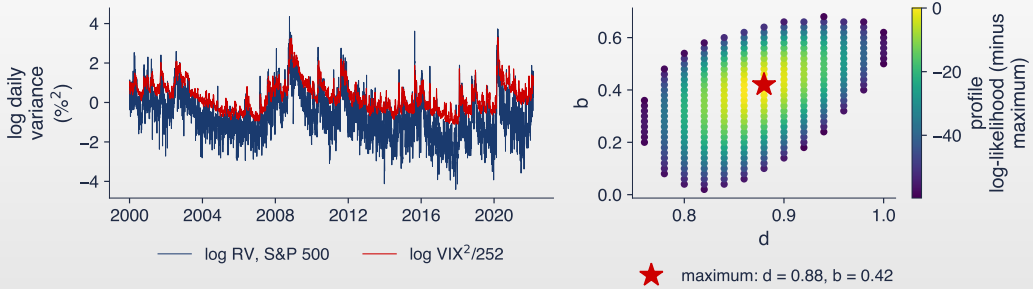
Semiparametric co-memory

- Narrow-band least squares (NBLS) Robinson (1994), Christensen and Nielsen (2006): regress y on x using only the m lowest frequencies

$$\hat{\beta} = \frac{\mathrm{Re} \sum_{j=1}^{m} I_{xy}(\lambda_j)}{\sum_{j=1}^{m} I_{xx}(\lambda_j)}$$

  - I_{xy}: cross-periodogram of x and y; I_{xx}: periodogram of x; Re: real part
  - only the low frequencies, where the common long-memory component lives: robust to short-run correlation between x and the error that biases OLS
- Rank by ELW Nielsen and Shimotsu (2007): estimate d of each series, test equal d, then count the small eigenvalues of the low-frequency spectral matrix
- Implied and realised variance Bandi and Perron (2006): does the VIX contain the long-memory component of future realised variance?
  - if $\log RV$ and $\log VIX^2$ share one fractional trend, a combination has lower d and the VIX forecasts the persistent part

FCVAR: realised and implied variance



- S&P 500 log 5-minute RV and $\log(\mathrm{VIX}^2/252)$, $n = 5\,523$ common days to 25 February 2022; right: profile log-likelihood of the rank-one FCVAR with $k = 0$ over (d, b), the 60 log-points below the maximum

ATS_ch10_fcvar

Interpreting the FCVAR

- Local Whittle: $d = 0.62$ for log RV, 0.83 for log VIX^2, 0.49 for the spread $\beta' X_t$ (SE 0.03): the combination is less persistent
- FCVAR: $\hat{d} = 0.88$, $\hat{b} = 0.42$, $\beta = (1, -1.70)$, $\alpha = (-1.55, 0.006)$: realised variance adjusts, the VIX does not (weak exogeneity)
- Trace tests ($\hat{b} < 1/2$, so χ^2): rank 0 against 2: 1453.94 (p <0.001); rank 1 against 2: 18.26 (p <0.001): with $k = 0$ the data also reject a single relation
- NBLS slope of log RV on log VIX^2: 1.27, against $1/1.70 = 0.59$ implied by FCVAR: short-run dynamics ($k > 0$) and the overnight gap of RV are left out; a project question

Recap: Fractional cointegration

- FCVAR generalises the VECM with two memory parameters d and b, estimated by profile likelihood and reduced-rank regression
- Rank tests are χ^2 only for $b < 1/2$; NBLS and ELW give semiparametric checks
- Realised and implied variance share persistent components; the exact structure needs short-run dynamics

FIGARCH (1/2): hyperbolic ARCH weights

- GARCH(1,1) as ARCH(∞): the weights on past squared shocks decay geometrically

$$\sigma_t^2 = \frac{\omega}{1-\beta} + \sum_{k\geq 1} \alpha\beta^{k-1}\varepsilon_{t-k}^2$$

- FIGARCH(1,d,1) Baillie, Bollerslev and Mikkelsen (1996): a fractional difference makes them decay hyperbolically

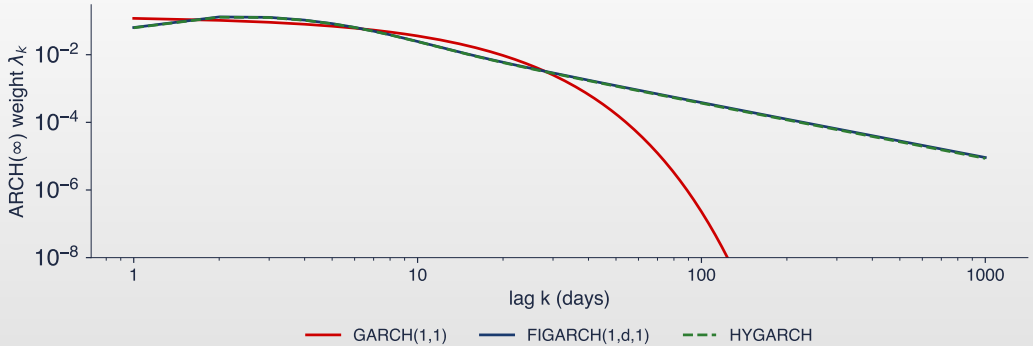
$$\sigma_t^2 = \frac{\omega}{1-\beta} + \Big[1 - \frac{(1-\phi L)(1-L)^d}{1-\beta L}\Big]\varepsilon_t^2 = \frac{\omega}{1-\beta} + \sum_{k\geq 1} \lambda_k \varepsilon_{t-k}^2$$

  - $d \in (0,1)$: memory of volatility; ϕ, β: short-run ARCH and GARCH terms; $\lambda_k \sim c\, k^{-d-1}$: the ARCH($\infty$) weights
  - positivity of all λ_k needs restrictions, e.g. $\beta - d \leq \phi \leq (2-d)/3$

FIGARCH (2/2): the variance paradox and HYGARCH

- Paradox: $\sum_k \lambda_k = 1$, so $\mathbb{E}\varepsilon_t^2 = \infty$
  - FIGARCH is not covariance stationary for any $d > 0$, yet strictly stationary
  - the long memory of ε_t^2 is therefore not defined by an autocovariance
- HYGARCH Davidson (2004): $(1-L)^d$ is replaced by $1 + a\big((1-L)^d - 1\big)$
  - $a \geq 0$: amplitude of the hyperbolic part; $a < 1$ gives a finite variance and hyperbolic memory, $a = 1$ is FIGARCH, $a = 0$ is GARCH

ARCH weights: geometric and hyperbolic



- S&P 500 daily returns 2000–2026 ($n = 6\,714$), Student t QML, ARCH(∞) truncated at 1000 lags; GARCH $(\alpha, \beta) = (0.119, 0.876)$, FIGARCH $d = 0.61$, $\phi = 0.04$, $\beta = 0.59$

ATS_ch10_figarch

Interpreting the FIGARCH estimates

Market	$\hat{d}$	LR	$\hat{a}$	BIC GARCH	BIC FIGARCH	weight after lag 22
S&P 500	0.61	40.8	0.99	18197	18165	15% / 5%
BET	0.46	79.0	1.00	19514	19444	17% / 1%
Bitcoin	0.98	13.6	1.06	20942	20936	12% / 8%

- LR of FIGARCH against GARCH (one more parameter): 40.8, 79.0, 13.6; BIC prefers FIGARCH in all three markets
- Weight on squared shocks older than 22 days: FIGARCH 15%, GARCH 5% (S&P 500); after 250 days 2.0% against practically zero
- HYGARCH amplitude $\hat{a} = 0.99$ and 1.00: no evidence against $a = 1$ (LR 0.22 and 0.01); Bitcoin $\hat{d} = 0.98$ sits at the integrated edge
- LR: likelihood-ratio statistic of FIGARCH against GARCH; $\hat{a}$: HYGARCH amplitude; last column: FIGARCH / GARCH

Long-memory stochastic volatility and noise

- LMSV Breidt, Crato and de Lima (1998): log variance is a long-memory process; squaring and taking logs makes the model linear

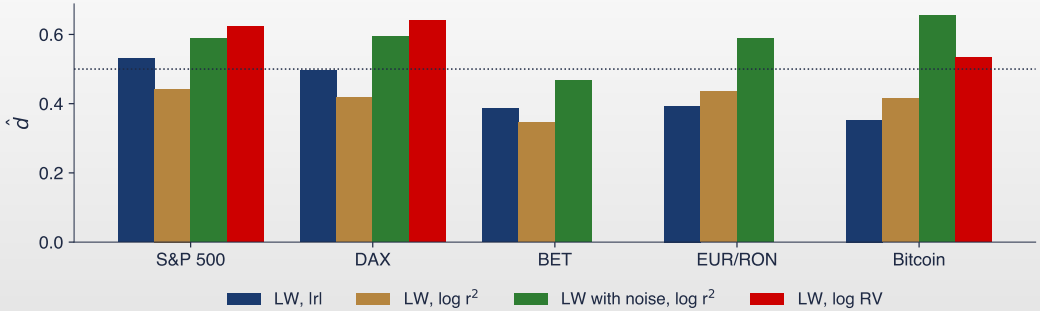
$$r_t = \sigma_t \eta_t, \quad \log \sigma_t^2 = \mu + h_t \;\Rightarrow\; \log r_t^2 = \mu + \mathbb{E} \log \eta^2 + h_t + \xi_t$$

  - h_t: ARFIMA(p, d, q); η_t: i.i.d. shock; $\xi_t = \log \eta_t^2 - \mathbb{E} \log \eta^2$: i.i.d. noise with variance $\pi^2/2$ for Gaussian η, often larger than $\mathrm{Var}(h_t)$
- Spectrum of $\log r_t^2$: a pole plus a flat noise floor

$$f(\lambda) = G \lambda^{-2d} + \frac{\sigma_\xi^2}{2\pi}$$

  - the flat noise flattens the low-frequency slope and biases $\hat{d}$ down
  - LW with noise Hurvich, Moulines and Soulier (2005): $f(\lambda) = G(\lambda^{-2d} + \theta)$, with the noise-to-signal ratio $\theta \geq 0$ estimated jointly; consistent, at a slower rate
- Realised variance reduces the noise by averaging intraday returns Andersen et al. (2003): $\log RV_t = \log IV_t +$ small error
- Continuous-time long memory in volatility: Comte and Renault (1998); the rough alternative follows in Section 6

Memory of volatility proxies across markets



- Local Whittle $\hat{d}$ with $m = n^{0.65}$ of $|r_t|$ and $\log(r_t^2 + c)$, $c = 1\%$ of the variance; LW with noise on $\log(r_t^2 + c)$ with $m = n^{0.8}$; log RV where realised measures exist (S&P 500, DAX to 2022; Bitcoin to 2026)

ATS_ch10_figarch

Interpreting volatility memory across markets

- S&P 500: $|r|$ 0.53, $\log r^2$ 0.44, with noise 0.59, log RV 0.62: the noise correction moves $\log r^2$ to the RV level
- Bitcoin: $\log r^2$ 0.41, with noise 0.66, log RV 0.53; DAX: 0.42, 0.59, 0.64
- Qu test on $|r_t|$ (1% value 1.56): S&P 500 0.78, DAX 0.72, Bitcoin 0.79; BET 1.71 and EUR/RON 1.88 reject
- For the BET and the leu, part of the apparent memory comes from regime changes (2008, the managed float): model the breaks first (Chapter 2)

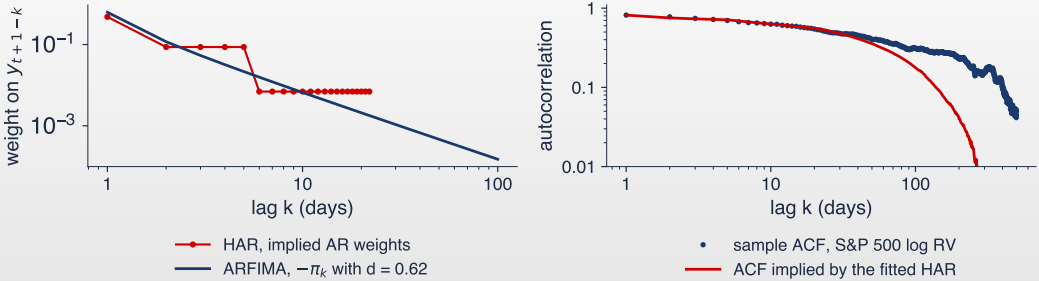
HAR as an approximation of long memory

- HAR Corsi (2009) on $y = \log RV$ (Chapter 8)

$$y_{t+1} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+1}$$

  - $\bar{y}_t^{(5)}$, $\bar{y}_t^{(22)}$: averages of y over the last 5 and 22 days; an AR(22) with three free slopes: a step-function approximation of hyperbolically decaying AR weights
- Formally short memory: its ACF decays geometrically, but over horizons up to a few months it mimics k^{2d-1}
  - cascade interpretation: traders with daily, weekly and monthly horizons (heterogeneous market hypothesis)
- Estimated by OLS, easy to extend (jumps, semivariances, Chapter 8): the benchmark every long-memory or rough model must beat

HAR weights and the ACF it implies



⊡ S&P 500 log RV; HAR by OLS: $\beta_d = 0.39$, $\beta_w = 0.40$, $\beta_m = 0.15$; left: implied AR weights and ARFIMA $-\pi_k$ with $d = 0.62$; right: ACF of a long simulation of the fitted HAR and the sample ACF

ATS_ch10_figarch

Interpreting the HAR approximation

- Sum of HAR weights 0.946: close to a unit root, which is how a short-memory model buys persistence
- ACF at lag 100: sample 0.31, HAR 0.179; at lag 250: 0.18 against 0.017
- The implied ACF falls below half the sample ACF after lag 112: HAR captures memory up to about half a year
- For forecasts up to a month this is enough, which is why HAR is hard to beat (Section 7)

Recap: Long memory in volatility

- FIGARCH has hyperbolic ARCH weights but infinite variance; HYGARCH nests it
- $\log r_t^2$ is long memory plus large noise: use the noise-robust LW or realised measures
- HAR is a short-memory model that mimics long memory over the horizons that matter for forecasting

Fractional Brownian motion (1/2)



A. N. Kolmogorov

Photo: Konrad Jacobs; CC BY-SA 2.0 de; Wikimedia Commons



B. B. Mandelbrot, TED 2010

Photo: Steve Jurvetson (2010); CC BY 2.0; Wikimedia Commons

- fBm W_t^H Mandelbrot and Van Ness (1968): a centred Gaussian process with $W_0^H = 0$ and covariance

$$\mathbb{E} W_t^H W_s^H = \frac{1}{2}\left(t^{2H} + s^{2H} - |t-s|^{2H}\right), \qquad H \in (0,1)$$

  - H: the Hurst exponent; $H = 1/2$ gives standard Brownian motion
  - stationary increments with $\mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$
  - self-similar: $W_{ct}^H \overset{d}{=} c^H W_t^H$ (equal in distribution) for $c > 0$

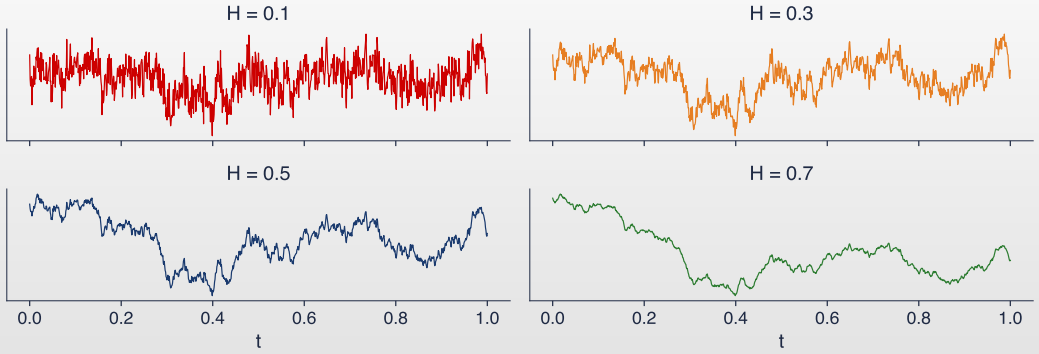
Fractional Brownian motion (2/2)

- Paths are Hölder continuous of every order $< H$
  - Hölder order: how fast $|W^H_{t+\Delta} - W^H_t|$ can shrink with Δ; $H < 1/2$: rougher (more irregular), $H > 1/2$: smoother than Brownian motion
- The increments, fractional Gaussian noise (fGn), have autocorrelations

$$\rho(k) = \frac{1}{2}\left(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H}\right) \sim H(2H-1)k^{2H-2}$$

  - long memory for $H > 1/2$ ($d = H - 1/2$); negative correlation for $H < 1/2$

Fractional Brownian paths



- One path each on 1000 steps, exact simulation by circulant embedding; $H = 0.5$ is Brownian motion

ATS_ch10_simulation

Interpreting the paths

- Lag-one correlation of the increments: -0.426 for $H = 0.1$, -0.242 for $H = 0.3$, 0 for $H = 0.5$, 0.320 for $H = 0.7$
- $H = 0.1$: every rise is likely to be followed by a fall, so the path is jagged at all scales but wanders little
- $H = 0.7$: trends persist; this is the fBm reading of the Hurst effect
- Roughness is a statement about small scales, persistence about large ones: keep the two apart

Simulating fBm: Cholesky, circulant embedding

- Cholesky: $\Gamma = LL'$, $X = LZ$; exact but $O(n^3)$ time and $O(n^2)$ memory
  - Γ: the $n \times n$ covariance matrix of the path; L: its lower-triangular Cholesky factor; Z: a vector of i.i.d. $N(0,1)$
- Davies–Harte Davies and Harte (1987), Dietrich and Newsam (1997): embed the Toeplitz Γ of fGn in a circulant matrix of order $M = 2(n-1)$ with first row $c = (\gamma_0, \ldots, \gamma_{n-1}, \gamma_{n-2}, \ldots, \gamma_1)$
  - γ_k: autocovariances of fGn; a circulant matrix is diagonalised by the discrete Fourier transform: eigenvalues $\Lambda = \mathrm{FFT}(c)$
  - if $\Lambda \geq 0$: $Y = \mathrm{FFT}\left(\sqrt{\Lambda/M}\,(Z_1 + iZ_2)\right)$, Z_1, Z_2 independent standard Gaussian vectors; $\mathrm{Re}\,Y$ and $\mathrm{Im}\,Y$ are two independent exact paths
- $O(n \log n)$; for fGn $\Lambda \geq 0$ for every H (Seminar 10, A7); fBm = cumulated fGn
- The same algorithm simulates ARFIMA$(0, d, 0)$ exactly (used in every Monte Carlo of this chapter)

The hybrid scheme for Volterra processes

- Rough volatility models use Volterra processes, not stationary-increment fBm Bayer, Friz and Gatheral (2016)

$$X_t = \int_0^t g(t-s)\, dW_s, \qquad g(x) = x^{H-1/2} \quad \text{(Riemann–Liouville)}$$

  - g: the kernel, singular at 0 when $H < 1/2$; W: Brownian motion; circulant embedding does not apply (non-stationary, possibly non-Gaussian when driven by a correlated price)

- Hybrid scheme Bennedsen, Lunde and Pakkanen (2017): integrate the kernel exactly on the last step, use a Riemann sum at optimal points further away

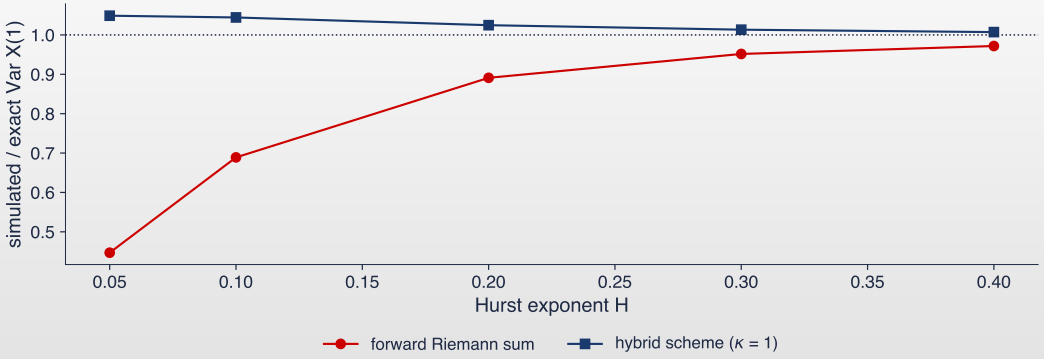
$$X_i \approx \int_{t_{i-1}}^{t_i} (t_i - s)^{H-1/2} dW_s + \sum_{k \geq 2} \Big(\frac{b_k}{n}\Big)^{H-1/2} \Delta W_{i-k+1}$$

$$b_k = \Big(\frac{k^{H+1/2} - (k-1)^{H+1/2}}{H+1/2}\Big)^{1/(H-1/2)}$$

  - n: steps per unit of time; ΔW_i: Brownian increments; b_k: the optimal evaluation point in cell k; the first term and ΔW_i are jointly Gaussian and simulated exactly

- The sum is a convolution: $O(n \log n)$ by FFT; the scheme extends to Brownian semistationary processes

Riemann sum against the hybrid scheme



⊡ Var$X(1)$ of the Riemann–Liouville process, exact value $1/(2H)$; $n = 250$ steps, 4 000 paths

ATS_ch10_simulation

Interpreting the simulation schemes

- Forward Riemann sum: 0.45 of the true variance at $H = 0.05$ and 0.69 at $H = 0.1$: it misses the mass of the kernel near the singularity
- Hybrid scheme: 1.05 and 1.04, within Monte Carlo error of 1
- The error of the naive scheme grows exactly where rough volatility lives ($H \approx 0.1$): option prices from it would be biased

The evidence: scaling of log volatility (1/2)

- Gatheral, Jaisson and Rosenbaum (2018), Section 2: from daily $\sigma_t = \sqrt{RV_t}$ (Oxford-Man library), the empirical moments of the increments of $\log \sigma$

$$m(q, \Delta) = \frac{1}{N} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q$$

  - Δ: lag in days; $q > 0$: order of the moment; N: number of pairs; $m(2, \Delta)$ is the variogram
- If $\log \sigma$ has fBm-like increments, the moments scale as a power of Δ (monofractal scaling)

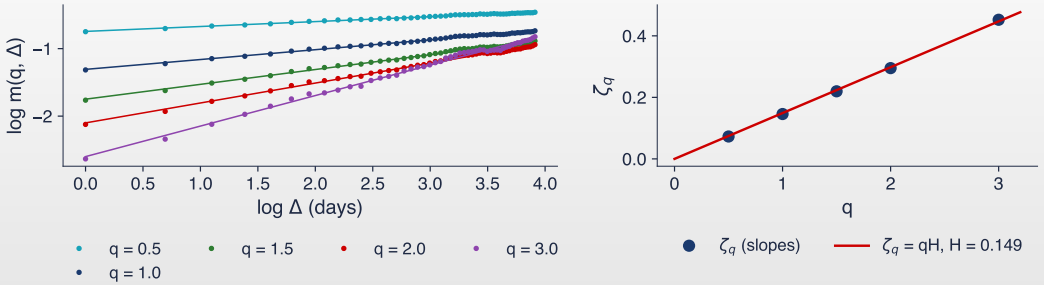
$$m(q, \Delta) = K_q \nu^q \Delta^{\zeta_q}, \qquad \zeta_q = qH$$

  - ν: volatility of volatility; $K_q = \mathbb{E}|Z|^q$, $Z \sim N(0,1)$; ζ_q: scaling exponent, linear in q with slope H
  - derivation and deviations from this scaling: Appendix

The evidence: scaling of log volatility (2/2)

- Procedure: OLS of $\log m(q, \Delta)$ on $\log \Delta$ for each q gives ζ_q; regress ζ_q on q through the origin to get H
  - we use $q \in \{0.5, 1, 1.5, 2, 3\}$ and $\Delta = 1, \ldots, 50$ days
- Their finding: linear ζ_q and H of order 0.1 for all indices; the increments of log volatility are close to Gaussian
- This contradicts every Markovian SV model (Heston, log-OU), whose log volatility has $H = 1/2$ at short lags

Replicating the scaling on the S&P 500



Oxford-Man 5-minute RV of the S&P 500, January 2000 – February 2022; left: $\log m(q, \Delta)$ and OLS lines; right: ζ_q and the line qH

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Interpreting the scaling

- $\zeta_q = 0.073, 0.146, 0.220, 0.295, 0.452$ for $q = 0.5, 1, 1.5, 2, 3$: linear in q, $\hat{H} = 0.149$
- Vol-of-vol $\hat{\nu} = 0.35$ from $m(2, \Delta) = \nu^2 \Delta^{2H}$; subsamples: 0.140 (2000–2010), 0.155 (2011–2022)
- The paper's result replicates: roughly $H \approx 0.1$ to 0.15, stable over time

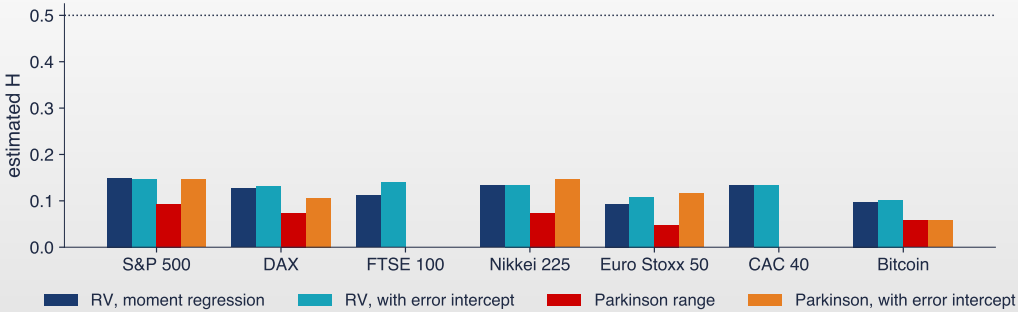
The RFSV model

- Gatheral, Jaisson and Rosenbaum (2018): rough fractional stochastic volatility (RFSV), log volatility a fractional Ornstein–Uhlenbeck (OU) process

$$\sigma_t = \exp(X_t), \qquad dX_t = \nu\, dW_t^H - \alpha(X_t - m)\, dt, \qquad H < 1/2$$

  - ν: volatility of volatility; m: long-run mean of X; $\alpha > 0$: speed of mean reversion, very small
  - for $\alpha T \ll 1$ (T: sample span) the increments behave like those of νW^H; stationarity only shows at horizons of order $1/\alpha$
- The link with long memory: over observable horizons $\log \sigma$ is close to a non-stationary fBm with $H \approx 0.1$, whose ARFIMA reading is $d = H + 1/2$ (0.65 for the S&P 500, close to the local Whittle $\hat{d}$ of log RV)
  - GJR show that local Whittle and similar estimators applied to RFSV simulations return the "long-memory" values found in the literature
- Pricing: rough Bergomi Bayer, Friz and Gatheral (2016) and rough Heston El Euch and Rosenbaum (2019)
  - both fit the term structure of the implied-volatility skew, $\propto \tau^{H-1/2}$ (τ: option maturity), with few parameters
  - option-based estimates also give small H Livieri et al. (2018)
  - the pricing side belongs to MFM; here we test the time-series claim

Roughness across markets and proxies



- *H* from 5-minute RV (Oxford-Man; Bitcoin: Binance) by the moment regression and with a measurement-error intercept; Parkinson range of EODHD daily highs and lows over the same days, where available

ATS_ch10_rough

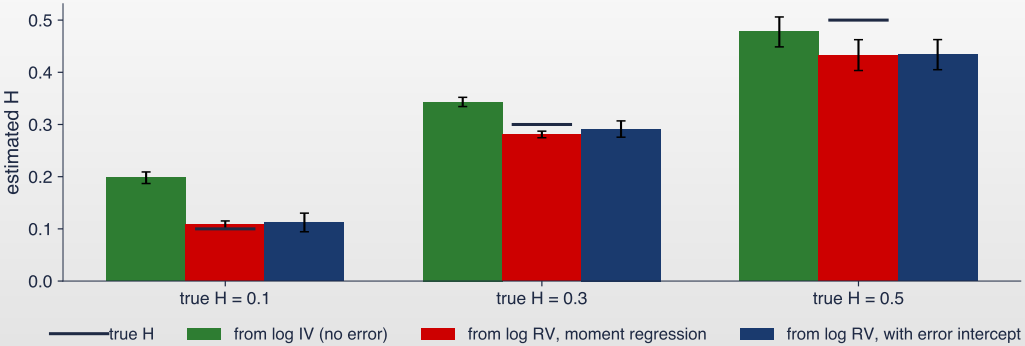
Interpreting roughness across markets

- RV: H from 0.093 to 0.149 for six indices and Bitcoin (S&P 500 0.149, DAX 0.127, Euro Stoxx 50 0.093, Bitcoin 0.098)
- The error intercept changes little (S&P 500 0.148); only for the FTSE 100 does it absorb 31% of $m(2,1)$
- Parkinson range, a much noisier proxy: S&P 500 0.09, DAX 0.07; with the intercept 0.15 and 0.11 (noise share 52% and 42%)
- Measurement error pushes H down; the correction brings the range back near the RV estimate, but not above it

Critiques: is roughness an artefact?

- RV is an estimate of integrated variance: $\log RV_t = \log \int_{t-1}^{t} \sigma_s^2 ds + \epsilon_t$
  - two opposite biases: daily integration smooths (pushes $\hat{H}$ up), the error ϵ_t adds a nugget to $m(2, \Delta)$ (pushes $\hat{H}$ down)
- Fukasawa, Takabatake and Westphal (2022): a Whittle-type estimator consistent under high-frequency asymptotics with measurement error; volatility remains rough, H even below 0.1
  - Bolko et al. (2023): GMM on the moments of integrated variance with noise; also small H
- Cont and Das (2024): the roughness estimators applied to RV can report small H even for smooth (Markovian) volatility: a finite-sample artefact of the proxy
  - the answer depends on the size of the error relative to the daily variation of log volatility: simulate it

Measurement error and the estimate of H



- Log volatility a fractional OU on 78 intraday steps a day, 2 500 days, 8 replications; daily IV (no error) and RV from the 78 squared returns; GJR estimator

ATS_ch10_rough

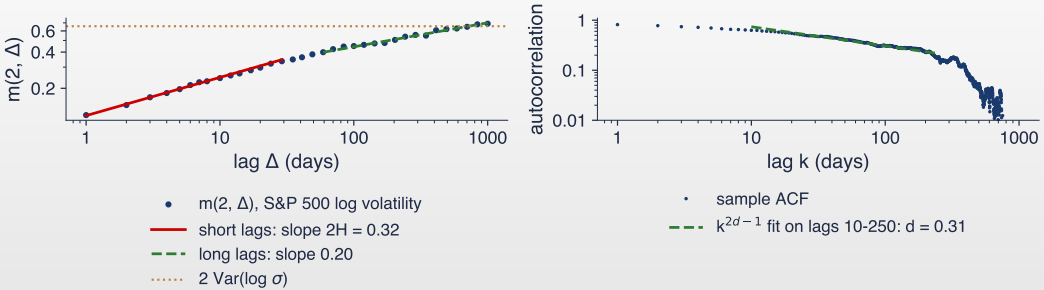
Interpreting the simulation

- True $H = 0.1$: from IV 0.20, from RV 0.11, RV with intercept 0.11: the integration bias and the error bias almost cancel
- True $H = 0.5$: IV 0.48, RV 0.43, with intercept 0.43; true $H = 0.3$: 0.34, 0.28, 0.29
- With 5-minute RV the error is too small to turn $H = 0.5$ into the 0.1–0.15 seen in the data
- A noisier proxy (Parkinson, squared returns) or a different volatility model could; the conclusion is conditional on this design

Roughness and persistence

- In fBm one parameter fixes both: small H means rough paths and (for increments) antipersistence
  - Cauchy and gamma-kernel models separate the fractal dimension (short scales) from the Hurst effect (long scales) Gneiting and Schlather (2004)
- Bennedsen, Lunde and Pakkanen (2022): Brownian semistationary model $X_t = \int_{-\infty}^{t} g(t-s)\, dW_s$, $g(x) \approx x^{\alpha}$ near 0 (roughness, $H = \alpha + 1/2$), $g(x) \approx x^{-\beta}$ at infinity (memory)
  - their finding: log volatility is both rough ($\alpha < 0$) and persistent ($\beta < 1$); the two are estimated separately
- Empirical check: slope of the variogram at short lags against the decay of the ACF at long lags

Short scales and long scales



S&P 500 log volatility, lags 1–1000 days; left: variogram $m(2, \Delta)$ with slopes on lags 1–10 and 100–1000; right: sample ACF with a k^{2d-1} fit on lags 10–250

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Interpreting the two scales

- Short lags: slope $2H = 0.32$, $H = 0.16$; long lags: slope 0.20, still rising at 1000 days, above $2\mathrm{Var}(\log \sigma)$
- Read as a non-stationary fractional process, the long-lag slope $2d - 1$ gives $d = 0.60$; local Whittle with $m = 74$ gives 0.56
- The ACF fit gives only $d = 0.31$: the sample ACF is biased down when d is near or above $1/2$
- Rough at short scales, highly persistent at long scales: both claims hold for the same series

Recap: Rough volatility

- fBm with $H < 1/2$: rough paths; simulate it exactly by circulant embedding, Volterra versions by the hybrid scheme
- The scaling of log RV gives $H \approx 0.1$–0.15 on every market we tried, stable over time
- Measurement error and daily integration bias $\hat{H}$ in opposite directions; roughness and persistence are separate properties

The RFSV predictor (1/2)

- For fBm (Nuzman–Poor), with $\alpha \approx 0$, Gatheral, Jaisson and Rosenbaum (2018), Section 5: the forecast of log variance is a weighted average of its past

$$\mathbb{E}[\log \sigma^2_{t+h} \mid \mathcal{F}_t] = \frac{\cos(H\pi)}{\pi} h^{H+1/2} \int_{-\infty}^{t} \frac{\log \sigma^2_s}{(t-s+h)(t-s)^{H+1/2}} \, ds$$

  - h: forecast horizon; $\mathcal{F}_t$: the past of volatility up to t; the kernel integrates to one, with more weight on the recent past for small H
  - discretised on days: $w_k = \int_k^{k+1} du/((u+h)u^{H+1/2})$, truncated at 500 days and normalised

The RFSV predictor (2/2): the variance forecast

- From log variance to variance: a lognormal correction

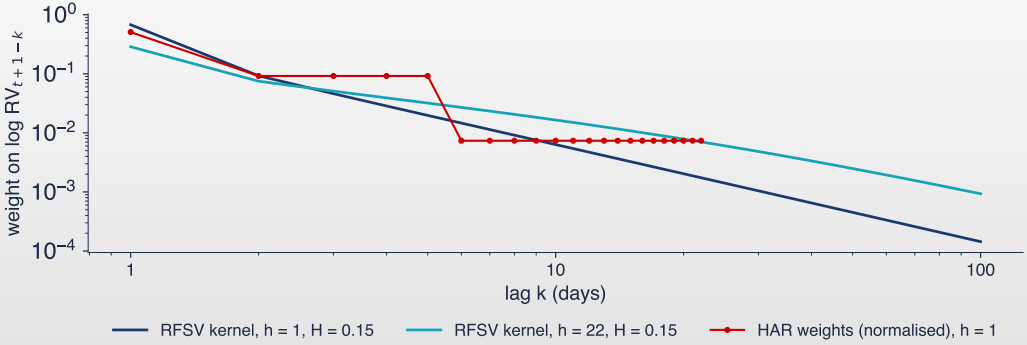
$$\mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t] = \exp\left(\mathbb{E}[\log \sigma^2_{t+h} \mid \mathcal{F}_t] + 2c\nu^2 h^{2H}\right)$$

$$c = \frac{\Gamma(3/2 - H)}{\Gamma(H + 1/2)\Gamma(2 - 2H)}$$

  - $2c\nu^2 h^{2H}$: half the conditional variance of $\log \sigma^2_{t+h}$, which grows with the horizon; with the S&P 500 values $c = 0.707$

- Two parameters (H, ν), both from the variogram: no likelihood, no optimisation

How far back the forecasts look



RFSV kernel with $H = 0.15$ (S&P 500) for $h = 1$ and $h = 22$ days, against the normalised HAR weights for $h = 1$

ATS_ch10_forecast

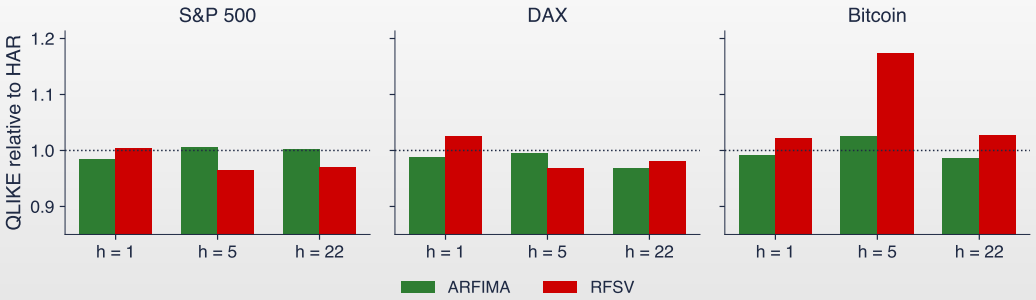
Interpreting the kernels

- $h = 1$: RFSV puts 68% on the last day, 86% on the last week, 95% on the last month; HAR: 51% and 87%
- $h = 22$: only 29% on the last day, 71% on the last month, 90% within 100 days: longer horizons look further back
- The kernel adapts its memory to the horizon automatically; HAR needs a separate regression for each horizon
- RFSV has no mean reversion: in long calm periods it does not pull forecasts towards a long-run level

A rigorous out-of-sample comparison

- Design fixed in advance: rolling window of 1000 days, parameters re-estimated every 20 days with data up to t only; target RV_{t+h}, $h \in \{1, 5, 22\}$
  - HAR: direct regression of $\log RV_{t+h}$; ARFIMA$(0, d, 0)$: local Whittle d (capped at 0.49), AR(∞) iterated; RFSV: H, ν from the window
- Loss: QLIKE $L = RV/F - \log(RV/F) - 1$, robust to the noise in the RV proxy Patton (2011); also MSE of logs
  - level forecasts from log forecasts need the lognormal correction: $\exp(\hat{\mu} + \hat{s}^2/2)$ for HAR and ARFIMA, $2c\nu^2 h^{2H}$ for RFSV
- Inference: DM with Newey–West variance (Chapter 1) Diebold and Mariano (1995); the loss differences are themselves persistent, so the p-values are optimistic

RFSV and ARFIMA against HAR



- Average QLIKE relative to HAR (below 1: better than HAR); a star marks a DM p-value below 5%; S&P 500 6 January 2004 – 24 January 2022 ($T = 4\,531$), DAX ($T = 4\,593$), Bitcoin 10 October 2020 – 27 August 2026 ($T = 2\,144$)

ATS_ch10_forecast

Interpreting the forecast comparison

QLIKE / HAR	S&P 500		DAX		Bitcoin	
	ARFIMA	RFSV	ARFIMA	RFSV	ARFIMA	RFSV
$h = 1$	0.984	1.003	0.988	1.026	0.992	1.022
$h = 5$	1.006	0.964	0.996	0.969	1.025	1.174
$h = 22$	1.002	0.969	0.968	0.982	0.986	1.027

- S&P 500 and DAX: RFSV is 1.8% to 3.6% better than HAR at 5 and 22 days but worse at 1 day; DM at 5 days: -1.78 (p 0.07) and -1.62 (p 0.11)
- Bitcoin: RFSV is worse at every horizon (1.174 at 5 days); rolling $\hat{H}$ ranges from 0.04 to 0.15, median 0.08
- ARFIMA is within 3.2% of HAR everywhere; of the 18 QLIKE comparisons, none is significant at 5%
- Roughness is a robust stylised fact; its forecasting gain over HAR is small and horizon-dependent, as in Gatheral, Jaisson and Rosenbaum (2018) and Wang, Xiao and Yu (2023)

Recap: Forecasting

- RFSV forecasts with a horizon-dependent power kernel and a lognormal correction, from two variogram parameters
- Out of sample, RFSV, ARFIMA and HAR are close; RFSV gains a few percent at weekly and monthly horizons for equity indices
- Pre-register the design, use QLIKE, and treat DM p-values with care under persistent loss differences

An open question

- Is the roughness of volatility a property of the volatility process, or of the way we measure it?
  - formal: H_0: $H \geq 0.3$ for integrated variance; tested with estimators that are consistent under measurement error, on pre-registered assets, measures and samples
  - falsified if the corrected estimates stay below 0.2 across sampling frequencies and markets
- Why it matters: rough models change option pricing and hedging; if roughness is an artefact, Markovian models suffice
  - literature to start from: Gatheral, Jaisson and Rosenbaum (2018), Fukasawa, Takabatake and Westphal (2022), Bolko et al. (2023), Cont and Das (2024), Bennedsen, Lunde and Pakkanen (2022)

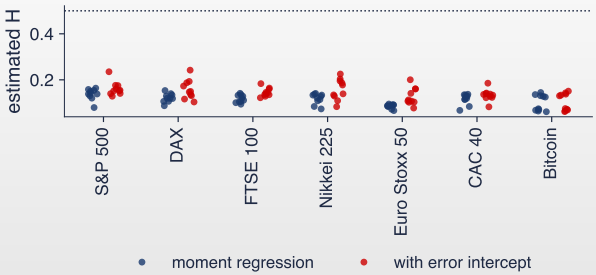
The discovery loop with an AI assistant

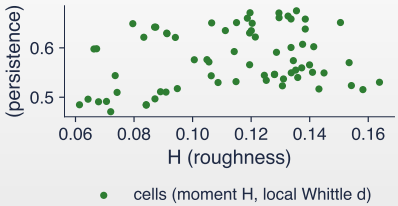
- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2018 that estimate the Hurst exponent of volatility with methods robust to measurement error; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Which data-generating processes with H = 0.5 would make the moment estimator on 5-minute RV return H near 0.1?`
  - **code and replication**: ask for the variogram regression, then reproduce first the S&P 500 value (0.149 here) and the simulation table
  - **critique**: `Act as a hostile referee: list how lag choice, overnight returns, jumps and microstructure noise could bias H.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The scale is stated: H of fBm, d of ARFIMA, $H = d + 1/2$ for stationary series; log volatility or log variance
- Bandwidths, lags and trimming are reported, with estimates across a range, not one convenient choice
- Breaks and level shifts are tested (Qu) before long memory is claimed
- Forecast comparisons are out of sample, pre-registered, with QLIKE and DM

Mini-case: how robust is "H of order 0.1"?





- 70 cells: seven assets × five realised measures × two halves of the sample; two estimators each (140 estimates); right: H against the local Whittle d of the same cell
- H from 0.06 to 0.24, median 0.13 (moments 0.12, with intercept 0.14); 96% below 0.2; d from 0.47 to 0.68; correlation of H and d 0.24: an AI summary that calls volatility "short memory because $H < 1/2$" is wrong

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Project idea

- **Roughness and persistence of volatility in Central and Eastern European markets**: replicate first, then extend
  - replicate: Gatheral, Jaisson and Rosenbaum (2018), Section 2 on the Oxford-Man S&P 500 ($H = 0.149$ here) and the forecasting comparison of their Section 5
  - extend: intraday data for the BET, WIG20 and EUR/RON; estimators robust to noise (FTW, GMM); Qu tests; a BSS model with separate roughness and memory
  - pre-register: assets, measures, lags, bandwidths, the forecast design and the loss
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Long memory is a pole at frequency zero; estimate it with local Whittle or exact local Whittle and report $\hat{d}(m)$
- Level shifts mimic long memory: test with Qu (2011) and look at the bandwidth signature
- Volatility is persistent over months (FIGARCH, LMSV, HAR) and rough over days ($H \approx 0.1$)
- Proxies matter: noise lowers $\hat{d}$ and $\hat{H}$; corrections exist and should be reported
- In forecasting, HAR remains a tough benchmark; rough models gain little and only at some horizons

Self-assessment

Questions

- Why is the asymptotic variance of local Whittle $1/(4m)$ and that of GPH $\pi^2/(24m)$?
- Why does local Whittle fail for $d > 1$, and how does exact local Whittle fix it?
- Which spectral shape does a random level shift produce?
- Why is FIGARCH not covariance stationary?
- How can a series be rough and long-memory at the same time?

Next:

- Spectral and wavelet analysis
- multitaper, coherence, wavelets: the frequency tools behind this chapter

Appendix: the variance of local Whittle

- Expand $R'(\hat{d}) = 0$ around d: $\sqrt{m}(\hat{d} - d) = -\sqrt{m}\, R'(d)/R''(\bar{d})$
- $\sqrt{m}\, R'(d) = \frac{2}{\sqrt{m}} \sum_j \nu_j(\xi_j - 1) + o_p(1)$, $\xi_j = \lambda_j^{2d} I(\lambda_j)/G$; with $\mathrm{Var}\xi_j \to 1$ and $\frac{1}{m}\sum \nu_j^2 \to 1$: $\to N(0, 4)$
- $R''(d) = \frac{4}{m} \sum_j \nu_j^2 \xi_j/\bar{\xi} - \left(\frac{2}{m} \sum_j \nu_j \xi_j/\bar{\xi}\right)^2 \to_p 4$
- Hence $\sqrt{m}(\hat{d} - d) \to N(0, 4/16) = N(0, 1/4)$; for GPH the same weights multiply $\log \xi_j$ with variance $\pi^2/6$: $N(0, \pi^2/24)$

Back

Appendix: scaling of fBm moments

- $W^H_{t+\Delta} - W^H_t \sim N(0, \Delta^{2H})$, so $\mathbb{E}|W^H_{t+\Delta} - W^H_t|^q = \Delta^{qH}\mathbb{E}|Z|^q$, $\mathbb{E}|Z|^q = 2^{q/2}\Gamma\left(\frac{q+1}{2}\right)/\sqrt{\pi}$
- With $\log \sigma_t = \nu W^H_t$: $\log m(q, \Delta) = q \log \nu + \log \mathbb{E}|Z|^q + qH \log \Delta$: slope $\zeta_q = qH$
- Departures: concave ζ_q signals multifractality or heavy tails; an intercept in $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$ signals i.i.d. measurement error of variance s^2
- fGn spectrum near zero: $f(\lambda) \propto |\lambda|^{1-2H}$, i.e. the increments are ARFIMA-like with $d = H - 1/2$

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References (1/4)

- Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625. doi:10.1111/1468-0262.00418
- Andrews, D. W. K., & Sun, Y. (2004). Adaptive local polynomial Whittle estimation of long-range dependence. *Econometrica*, 72(2), 569–614. doi:10.1111/j.1468-0262.2004.00501.x
- Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30. doi:10.1016/S0304-4076(95)01749-6
- Baillie, R. T., Chung, C.-F., & Tieslau, M. A. (1996). Analysing inflation by the fractionally integrated ARFIMA–GARCH model. *Journal of Applied Econometrics*, 11(1), 23–40. doi:10.1002/(SICI)1099-1255(199601)11:1<23::AID-JAE374>3.0.CO;2-M
- Bandi, F. M., & Perron, B. (2006). Long memory and the relation between implied and realized volatility. *Journal of Financial Econometrics*, 4(4), 636–670. doi:10.1093/jjfinec/nbl003
- Bayer, C., Friz, P., & Gatheral, J. (2016). Pricing under rough volatility. *Quantitative Finance*, 16(6), 887–904. doi:10.1080/14697688.2015.1099717
- Bennedsen, M., Lunde, A., & Pakkanen, M. S. (2022). Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5), 961–1006. doi:10.1093/jjfinec/nbaa049
- Bennedsen, M., Lunde, A., & Pakkanen, M. S. (2017). Hybrid scheme for Brownian semistationary processes. *Finance and Stochastics*, 21(4), 931–965. doi:10.1007/s00780-017-0335-5
- Beran, J., Feng, Y., Ghosh, S., & Kulik, R. (2013). *Long-Memory Processes: Probabilistic Properties and Statistical Methods*. Springer. doi:10.1007/978-3-642-35512-7
- Bolko, A. E., Christensen, K., Pakkanen, M. S., & Veliyev, B. (2023). A GMM approach to estimate the roughness of stochastic volatility. *Journal of Econometrics*, 235(2), 745–778. doi:10.1016/j.jeconom.2022.06.009
- Breidt, F. J., Crato, N., & de Lima, P. (1998). The detection and estimation of long memory in stochastic volatility. *Journal of Econometrics*, 83(1–2), 325–348. doi:10.1016/S0304-4076(97)00072-9
- Christensen, B. J., & Nielsen, M. Ø. (2006). Asymptotic normality of narrow-band least squares in the stationary fractional cointegration model and volatility forecasting. *Journal of Econometrics*, 133(1), 343–371. doi:10.1016/j.jeconom.2005.03.018
- Comte, F., & Renault, E. (1998). Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4), 291–323. doi:10.1111/1467-9965.00057
- Cont, R., & Das, P. (2024). Rough volatility: Fact or artefact? *Sankhya B*, 86(1), 191–223. doi:10.1007/s13571-024-00322-2

References (2/4)

- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- Davidson, J. (2004). Moment and memory properties of linear conditional heteroscedasticity models, and a new model. *Journal of Business & Economic Statistics*, 22(1), 16–29. doi:10.1198/073500103288619359
- Davies, R. B., & Harte, D. S. (1987). Tests for Hurst effect. *Biometrika*, 74(1), 95–101. doi:10.1093/biomet/74.1.95
- Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Dietrich, C. R., & Newsam, G. N. (1997). Fast and exact simulation of stationary Gaussian processes through circulant embedding of the covariance matrix. *SIAM Journal on Scientific Computing*, 18(4), 1088–1107. doi:10.1137/S1064827592240555
- El Euch, O., & Rosenbaum, M. (2019). The characteristic function of rough Heston models. *Mathematical Finance*, 29(1), 3–38. doi:10.1111/mafi.12173
- Fox, R., & Taqqu, M. S. (1986). Large-sample properties of parameter estimates for strongly dependent stationary Gaussian time series. *The Annals of Statistics*, 14(2), 517–532. doi:10.1214/aos/1176349936
- Fukasawa, M., Takabatake, T., & Westphal, R. (2022). Consistent estimation for fractional stochastic volatility model under high-frequency asymptotics. *Mathematical Finance*, 32(4), 1086–1132. doi:10.1111/mafi.12354
- Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949. doi:10.1080/14697688.2017.1393551
- Geweke, J., & Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238. doi:10.1111/j.1467-9892.1983.tb00371.x
- Gneiting, T., & Schlather, M. (2004). Stochastic models that separate fractal dimension and the Hurst effect. *SIAM Review*, 46(2), 269–282. doi:10.1137/S0036144501394387
- Granger, C. W. J., & Hyung, N. (2004). Occasional structural breaks and long memory with an application to the S&P 500 absolute stock returns. *Journal of Empirical Finance*, 11(3), 399–421. doi:10.1016/j.jempfin.2003.03.001
- Granger, C. W. J., & Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29. doi:10.1111/j.1467-9892.1980.tb00297.x

References (3/4)

- Granger, C. W. J. (1980). Long memory relationships and the aggregation of dynamic models. *Journal of Econometrics*, 14(2), 227–238. doi:10.1016/0304-4076(80)90092-5
- Hassler, U., & Wolters, J. (1995). Long memory in inflation rates: International evidence. *Journal of Business & Economic Statistics*, 13(1), 37–45. doi:10.1080/07350015.1995.10524577
- Heber, G., Lunde, A., Shephard, N., & Sheppard, K. K. (2009). Oxford-Man Institutes realized library, version 0.3. Oxford-Man Institute, University of Oxford. web.archive.org
- Henry, M., & Robinson, P. M. (1996). Bandwidth choice in Gaussian semiparametric estimation of long range dependence. In *Athens Conference on Applied Probability and Time Series Analysis*, Lecture Notes in Statistics 115, 220–232. Springer. doi:10.1007/978-1-4612-2412-9_16
- Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165
- Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799. doi:10.1061/TACEAT.0006518
- Hurvich, C. M., Deo, R., & Brodsky, J. (1998). The mean squared error of Geweke and Porter-Hudaks estimator of the memory parameter of a long-memory time series. *Journal of Time Series Analysis*, 19(1), 19–46. doi:10.1111/1467-9892.00075
- Hurvich, C. M., Moulines, E., & Soulier, P. (2005). Estimating long memory in volatility. *Econometrica*, 73(4), 1283–1328. doi:10.1111/j.1468-0262.2005.00616.x
- Johansen, S., & Nielsen, M. Ø. (2012). Likelihood inference for a fractionally cointegrated vector autoregressive model. *Econometrica*, 80(6), 2667–2732. doi:10.3982/ECTA9299
- Livieri, G., Mouti, S., Pallavicini, A., & Rosenbaum, M. (2018). Rough volatility: Evidence from option prices. *IISE Transactions*, 50(9), 767–776. doi:10.1080/24725854.2018.1444297
- Lu, Y. K., & Perron, P. (2010). Modeling and forecasting stock return volatility using a random level shift model. *Journal of Empirical Finance*, 17(1), 138–156. doi:10.1016/j.jempfin.2009.10.001
- MacKinnon, J. G., & Nielsen, M. Ø. (2014). Numerical distribution functions of fractional unit root and cointegration tests. *Journal of Applied Econometrics*, 29(1), 161–171. doi:10.1002/jae.2295
- Mandelbrot, B. B., & Van Ness, J. W. (1968). Fractional Brownian motions, fractional noises and applications. *SIAM Review*, 10(4), 422–437. doi:10.1137/1010093
- Nielsen, M. Ø., & Shimotsu, K. (2007). Determining the cointegrating rank in nonstationary fractional systems by the exact local Whittle approach. *Journal of Econometrics*, 141(2), 574–596. doi:10.1016/j.jeconom.2006.10.008

References (4/4)

- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *The Journal of Business*, 53(1), 61–65. doi:10.1086/296071
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- Perron, P., & Qu, Z. (2010). Long-memory and level shifts in the volatility of stock market return indices. *Journal of Business & Economic Statistics*, 28(2), 275–290. doi:10.1198/jbes.2009.06171
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Phillips, P. C. B., & Shimotsu, K. (2004). Local Whittle estimation in nonstationary and unit root cases. *The Annals of Statistics*, 32(2), 656–692. doi:10.1214/009053604000000139
- Qu, Z. (2011). A test against spurious long memory. *Journal of Business & Economic Statistics*, 29(3), 423–438. doi:10.1198/jbes.2010.09153
- Robinson, P. M. (1995a). Log-periodogram regression of time series with long range dependence. *The Annals of Statistics*, 23(3), 1048–1072. doi:10.1214/aos/1176324636
- Robinson, P. M. (1995b). Gaussian semiparametric estimation of long range dependence. *The Annals of Statistics*, 23(5), 1630–1661. doi:10.1214/aos/1176324317
- Robinson, P. M. (1994). Semiparametric analysis of long-memory time series. *The Annals of Statistics*, 22(1), 515–539. doi:10.1214/aos/1176325382
- Shimotsu, K. (2010). Exact local Whittle estimation of fractional integration with unknown mean and time trend. *Econometric Theory*, 26(2), 501–540. doi:10.1017/S0266466609100075
- Shimotsu, K., & Phillips, P. C. B. (2005). Exact local Whittle estimation of fractional integration. *The Annals of Statistics*, 33(4), 1890–1933. doi:10.1214/009053605000000309
- Velasco, C. (1999). Gaussian semiparametric estimation of non-stationary time series. *Journal of Time Series Analysis*, 20(1), 87–127. doi:10.1111/1467-9892.00127
- Wang, X., Xiao, W., & Yu, J. (2023). Modeling and forecasting realized volatility with the fractional Ornstein–Uhlenbeck process. *Journal of Econometrics*, 232(2), 389–415. doi:10.1016/j.jeconom.2021.08.001